

Machine Learning Practical

Viva Voce — Questions & Answers

Lab 01 — Python Libraries for ML (NumPy, Pandas, Matplotlib, Sklearn)

Q1. What is NumPy and why is it used in ML?

Ans: NumPy (Numerical Python) is a library for numerical computing. It provides fast multi-dimensional arrays (ndarray) and mathematical functions. In ML, it is used for matrix operations, dataset manipulation, and efficient numerical computations.

Q2. What is a NumPy ndarray?

Ans: An ndarray is an N-dimensional array object in NumPy. It stores elements of the same data type in a contiguous block of memory, allowing fast vectorized operations without Python loops.

Q3. What is Pandas and how is it used in ML?

Ans: Pandas provides DataFrame and Series structures for data manipulation. In ML, it is used to load datasets (CSV, Excel), handle missing values, filter/group data, and prepare features before training.

Q4. What is the difference between a Series and DataFrame?

Ans: A Series is a 1-dimensional labeled array. A DataFrame is a 2-dimensional labeled table with rows and columns, similar to a spreadsheet or SQL table.

Q5. What is Matplotlib?

Ans: Matplotlib is a plotting library for Python. It is used to visualize data through line plots, scatter plots, histograms, bar charts, etc., helping to understand data distributions and model performance.

Q6. What is Scikit-learn (sklearn)?

Ans: Scikit-learn is a Python ML library that provides tools for classification, regression, clustering, model selection, preprocessing, and evaluation. It has a consistent API for all algorithms.

Q7. What does train_test_split do?

Ans: It splits a dataset into training and testing subsets. The model learns from training data and is evaluated on unseen test data to measure generalization.

Lab 02 — FIND-S Algorithm

Q1. What is the FIND-S algorithm?

Ans: FIND-S is a concept learning algorithm that finds the most specific hypothesis consistent with all positive training examples. It starts with the most specific hypothesis and generalizes it whenever a positive example is misclassified.

Q2. What type of learning does FIND-S perform?

Ans: FIND-S performs supervised concept learning. It works only on positive examples and ignores negative examples.

Q3. What is a hypothesis in FIND-S?

Ans: A hypothesis is a conjunction of attribute constraints. '?' means any value is accepted, a specific value means only that value is accepted, and 'null/empty' means no value is accepted.

Q4. What are the limitations of FIND-S?

Ans: It ignores negative examples, cannot detect inconsistent data, finds only one hypothesis (the most specific), and cannot handle noisy data.

Q5. What is the most specific hypothesis initially?

Ans: The initial hypothesis is the most specific: it sets all attributes to the null/empty set ($\emptyset$), meaning it classifies nothing as positive.

Q6. How does FIND-S generalize a hypothesis?

Ans: When a positive example is not covered by the current hypothesis, each attribute is generalized: if the attribute value differs from the hypothesis, it is replaced with '?' (any value).

Lab 03 — Linear Regression

Q1. What is Linear Regression?

Ans: Linear Regression is a supervised learning algorithm that models the relationship between a dependent variable (y) and one or more independent variables (x) by fitting a straight line: $y = mx + b$.

Q2. What is the cost function in Linear Regression?

Ans: The cost function is Mean Squared Error (MSE): $MSE = (1/n) * \text{sum of } (y_{\text{actual}} - y_{\text{predicted}})^2$. The goal is to minimize this error.

Q3. What is Gradient Descent?

Ans: Gradient Descent is an optimization algorithm that iteratively adjusts model parameters (slope, intercept) in the direction that minimizes the cost function by computing gradients.

Q4. What is R-squared (R^2) score?

Ans: R^2 score measures how well the regression line fits the data. $R^2 = 1$ means perfect fit; $R^2 = 0$ means the model is no better than predicting the mean. It ranges from 0 to 1 (or negative for bad models).

Q5. What is the difference between Simple and Multiple Linear Regression?

Ans: Simple Linear Regression has one independent variable ($y = mx + b$). Multiple Linear Regression has two or more independent variables ($y = b_0 + b_1x_1 + b_2x_2 + \dots$).

Q6. What assumptions does Linear Regression make?

Ans: Linearity between variables, independence of errors, homoscedasticity (constant variance of errors), normality of residuals, and no multicollinearity among features.

Q7. What is overfitting in regression?

Ans: Overfitting occurs when a model fits the training data too well (including noise) but performs poorly on new/unseen data. It means the model has learned noise rather than the true pattern.

Lab 04 — Logistic Regression

Q1. What is Logistic Regression?

Ans: Logistic Regression is a supervised classification algorithm that predicts the probability of a binary outcome using the sigmoid function. Despite its name, it is used for classification, not regression.

Q2. What is the sigmoid function?

Ans: The sigmoid function maps any real value to a value between 0 and 1: $\text{sigma}(z) = 1 / (1 + e^{(-z)})$. It is used to convert the linear output into a probability.

Q3. What is the decision boundary in Logistic Regression?

Ans: The decision boundary is the threshold (usually 0.5) that separates two classes. If predicted probability ≥ 0.5 , predict class 1; otherwise predict class 0.

Q4. What is the cost function for Logistic Regression?

Ans: Log Loss (Binary Cross-Entropy): $\text{Cost} = -[y \cdot \log(h) + (1-y) \cdot \log(1-h)]$, where h is the predicted probability. This penalizes confident wrong predictions heavily.

Q5. What is the difference between Linear and Logistic Regression?

Ans: Linear Regression predicts continuous values; Logistic Regression predicts class probabilities. Linear uses MSE loss; Logistic uses Log Loss. Linear output is unbounded; Logistic output is between 0 and 1.

Q6. What metrics are used to evaluate a classification model?

Ans: Accuracy, Precision, Recall, F1-Score, AUC-ROC curve, and Confusion Matrix.

Q7. What is a confusion matrix?

Ans: A confusion matrix is a table showing True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). It helps evaluate classification performance.

Lab 05 — Candidate Elimination Algorithm

Q1. What is the Candidate Elimination Algorithm?

Ans: Candidate Elimination is a concept learning algorithm that maintains two boundary sets: the most specific hypotheses (S boundary) and the most general hypotheses (G boundary). It finds all consistent hypotheses.

Q2. What is the Version Space?

Ans: The Version Space is the set of all hypotheses consistent with the training examples. It lies between the S boundary (most specific) and G boundary (most general).

Q3. How does the algorithm handle a positive example?

Ans: For a positive example: generalize all hypotheses in S that do not cover it, remove hypotheses in G that do not cover it, and remove overly general hypotheses from S.

Q4. How does the algorithm handle a negative example?

Ans: For a negative example: specialize all hypotheses in G that cover it, remove hypotheses in S that cover it, and remove overly specific hypotheses from G.

Q5. What is the difference between FIND-S and Candidate Elimination?

Ans: FIND-S finds only the most specific hypothesis and ignores negative examples. Candidate Elimination finds the complete version space using both positive and negative examples.

Q6. When does learning succeed in Candidate Elimination?

Ans: Learning succeeds when S and G boundaries converge to a single hypothesis, meaning a unique concept has been learned from the data.

Lab 06 — Naive Bayes Classifier

Q1. What is the Naive Bayes classifier?

Ans: Naive Bayes is a probabilistic classifier based on Bayes' Theorem. It assumes all features are conditionally independent given the class label — this is the 'naive' assumption.

Q2. State Bayes' Theorem.

Ans: $P(\text{class} \mid \text{features}) = [P(\text{features} \mid \text{class}) * P(\text{class})] / P(\text{features})$. The class with the highest posterior probability is selected as the prediction.

Q3. What is the 'naive' assumption?

Ans: The naive assumption is that all features are independent of each other given the class. This simplifies computation but is rarely true in real data.

Q4. What are the types of Naive Bayes classifiers?

Ans: Gaussian Naive Bayes (for continuous data), Multinomial Naive Bayes (for word counts/text), and Bernoulli Naive Bayes (for binary features).

Q5. What is Laplace Smoothing?

Ans: Laplace Smoothing (add-1 smoothing) adds 1 to every feature count to avoid zero probability when a feature value is absent in training data. Zero probability would nullify the entire product.

Q6. What are the advantages of Naive Bayes?

Ans: It is simple, fast, works well with small data, handles high-dimensional data well, and performs well for text classification tasks like spam detection.

Q7. Where is Naive Bayes commonly used?

Ans: Email spam filtering, sentiment analysis, document classification, medical diagnosis, and recommendation systems.

Lab 07 — Decision Tree (ID3 Algorithm)

Q1. What is a Decision Tree?

Ans: A Decision Tree is a supervised learning algorithm that splits data into branches based on feature conditions, forming a tree structure where leaf nodes represent class labels or values.

Q2. What is the ID3 algorithm?

Ans: ID3 (Iterative Dichotomiser 3) builds a decision tree by selecting the attribute with the highest Information Gain at each step as the splitting criterion.

Q3. What is Entropy?

Ans: Entropy measures impurity or disorder in a dataset: $\text{Entropy} = -\sum(p_i \cdot \log_2(p_i))$. Entropy = 0 means all samples belong to one class (pure); Entropy = 1 means equal distribution.

Q4. What is Information Gain?

Ans: Information Gain = Entropy(parent) - weighted average Entropy(children). It measures how much an attribute reduces uncertainty. The attribute with the highest IG is chosen for splitting.

Q5. What is Gini Impurity?

Ans: Gini Impurity = $1 - \sum(p_i^2)$. It is another measure of impurity used in CART algorithm. Lower Gini means purer node. Gini = 0 means perfect purity.

Q6. What is pruning in Decision Trees?

Ans: Pruning reduces the size of a tree by removing branches that have little predictive power, preventing overfitting. Pre-pruning stops growth early; post-pruning removes branches after full growth.

Q7. What is the difference between ID3, C4.5, and CART?

Ans: ID3 uses Information Gain (categorical only). C4.5 uses Gain Ratio (handles numerical data). CART uses Gini Impurity and creates binary splits.

Lab 08 — Decision Tree on Iris Dataset

Q1. What is the Iris dataset?

Ans: The Iris dataset contains 150 samples of iris flowers with 4 features (sepal length, sepal width, petal length, petal width) and 3 classes: Setosa, Versicolor, Virginica.

Q2. Why is the Iris dataset popular in ML?

Ans: It is small, clean, well-balanced, and a good benchmark for testing classification algorithms. It was introduced by statistician Ronald Fisher in 1936.

Q3. What is max_depth in Decision Tree?

Ans: max_depth controls how deep the tree can grow. A small depth prevents overfitting; a very large depth may cause overfitting by memorizing training data.

Q4. How do you visualize a Decision Tree?

Ans: Using sklearn's plot_tree() or export_graphviz() functions, which show each split condition, Gini/entropy values, sample count, and class distribution at each node.

Q5. What does feature_importances_ tell you?

Ans: It shows the relative importance of each feature in making splits. Features with higher values contributed more to reducing impurity across the tree.

Lab 09 — Naive Bayes on Custom Dataset

Q1. How do you apply Naive Bayes to a custom CSV dataset?

Ans: Load data with Pandas, separate features (X) and target (y), split into train/test sets, fit GaussianNB() or MultinomialNB(), predict on test set, and evaluate accuracy.

Q2. What is GaussianNB?

Ans: GaussianNB assumes features follow a Gaussian (normal) distribution. It estimates mean and variance for each feature per class and uses the Gaussian PDF for probability calculation.

Q3. What preprocessing is needed before Naive Bayes?

Ans: Handling missing values, encoding categorical variables (LabelEncoder or OneHotEncoder), and for GaussianNB, features should ideally be normally distributed.

Q4. Can Naive Bayes handle missing values?

Ans: Naive Bayes can handle missing values by simply ignoring that feature in the probability calculation for a given instance.

Q5. What does classification_report show?

Ans: It shows Precision, Recall, F1-Score, and Support for each class. Precision = $TP/(TP+FP)$; Recall = $TP/(TP+FN)$; F1 = harmonic mean of Precision and Recall.

Lab 10 — KNN Algorithm

Q1. What is K-Nearest Neighbors (KNN)?

Ans: KNN is a non-parametric, lazy learning algorithm that classifies a new point based on the majority class of its K nearest neighbors in the training dataset.

Q2. What distance metrics are used in KNN?

Ans: Euclidean distance (most common): $\sqrt{\sum (x1-x2)^2}$, Manhattan distance: $\sum |x1-x2|$, and Minkowski distance (generalized form).

Q3. How do you choose the value of K?

Ans: K is chosen by cross-validation. Small K = low bias, high variance (overfitting). Large K = high bias, low variance (underfitting). Odd K avoids ties in binary classification.

Q4. Why is KNN called a lazy learner?

Ans: KNN does not build a model during training. It stores all training data and performs computation only at prediction time, making training fast but prediction slow for large datasets.

Q5. What is the effect of feature scaling on KNN?

Ans: KNN is highly sensitive to feature scales because it uses distance measures. Features with larger ranges dominate the distance. Normalization or Standardization (Min-Max or Z-score) is essential.

Q6. What are the disadvantages of KNN?

Ans: Slow prediction for large datasets ($O(n)$ per query), sensitive to irrelevant features and noise, requires feature scaling, and high memory usage (stores all training data).

Q7. What is the Curse of Dimensionality?

Ans: In high-dimensional spaces, all points become equidistant, making KNN ineffective. Distance metrics lose meaning as the number of features increases significantly.

Lab 11 — ANN for Classification

Q1. What is an Artificial Neural Network (ANN)?

Ans: An ANN is a computational model inspired by the human brain, consisting of layers of interconnected nodes (neurons). Each connection has a weight that is learned during training.

Q2. What are the layers in an ANN?

Ans: Input Layer (receives features), one or more Hidden Layers (learn representations), and Output Layer (produces predictions).

Q3. What is an activation function? Name some.

Ans: An activation function introduces non-linearity to the network. Common ones: ReLU ($\max(0, x)$) for hidden layers, Sigmoid (0 to 1) for binary output, Softmax for multi-class output, Tanh (-1 to 1).

Q4. What is Backpropagation?

Ans: Backpropagation is the training algorithm for ANNs. It computes the gradient of the loss function with respect to each weight using the chain rule, and updates weights to minimize the loss.

Q5. What is an epoch?

Ans: An epoch is one complete pass through the entire training dataset during training. Multiple epochs are needed to allow the network to learn the patterns.

Q6. What is the difference between batch, mini-batch, and stochastic gradient descent?

Ans: Batch GD: updates weights after all samples. Stochastic GD: updates after each sample. Mini-batch GD: updates after a small batch — best trade-off between speed and stability.

Q7. What is the vanishing gradient problem?

Ans: In deep networks with sigmoid/tanh, gradients become very small as they propagate backward, causing weights in early layers to barely update. ReLU activation largely solves this.

Q8. What is a Multi-Layer Perceptron (MLP)?

Ans: MLP is a type of feedforward ANN with one or more hidden layers. All layers are fully connected. It is trained using backpropagation and is the most common ANN architecture.

Lab 12 — K-Means Clustering

Q1. What is K-Means Clustering?

Ans: K-Means is an unsupervised clustering algorithm that partitions n data points into K clusters. Each point belongs to the cluster with the nearest centroid (cluster center).

Q2. Explain the steps of K-Means algorithm.

Ans: 1) Choose K initial centroids randomly. 2) Assign each point to the nearest centroid. 3) Recompute centroids as the mean of assigned points. 4) Repeat steps 2-3 until centroids stop changing.

Q3. What is the objective function of K-Means?

Ans: Minimize the Within-Cluster Sum of Squares (WCSS): sum of squared distances from each point to its cluster centroid. Also called inertia.

Q4. What is the Elbow Method?

Ans: Plot WCSS vs. K values. The 'elbow point' — where WCSS decreases sharply then levels off — indicates the optimal number of clusters.

Q5. What are the limitations of K-Means?

Ans: Must specify K in advance, sensitive to initial centroid placement, assumes spherical clusters of equal size, affected by outliers, and does not work well for non-convex shapes.

Q6. What is the difference between supervised and unsupervised learning?

Ans: Supervised learning uses labeled data to learn a mapping from input to output. Unsupervised learning finds hidden patterns or groupings in data without labels. K-Means is unsupervised.

Q7. What is Silhouette Score?

Ans: Silhouette Score measures clustering quality. It ranges from -1 to +1. Values near +1 mean points are well-matched to their own cluster and far from neighboring clusters. Values near 0 mean overlapping clusters.

Q8. What is K-Means++ initialization?

Ans: K-Means++ is an improved initialization method that spreads initial centroids far apart, leading to better convergence and reducing the chance of poor clustering due to random initialization.